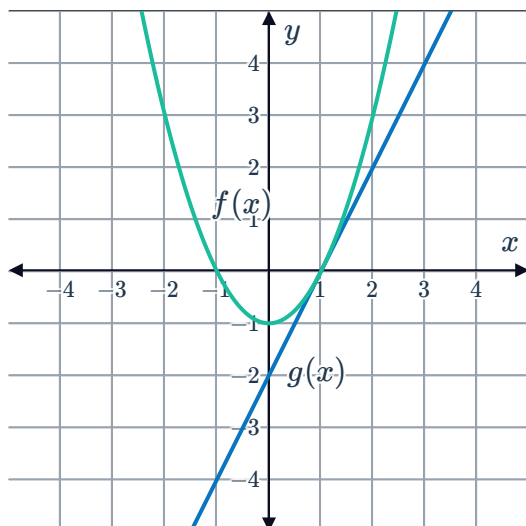


Equation of a tangent from a graph

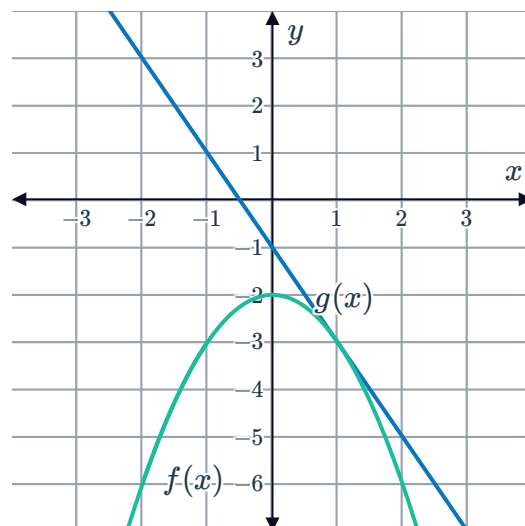


- 1 Each of the following graphs contain a curve, $f(x)$, along with one of its tangents, $g(x)$.
- State the coordinates of the point at which $g(x)$ is a tangent to the curve $f(x)$.
 - State the gradient of the tangent.
 - Hence determine the equation of the line $y = g(x)$.

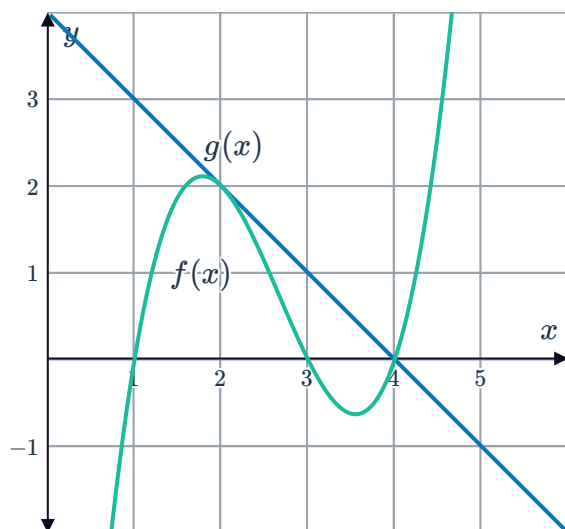
a



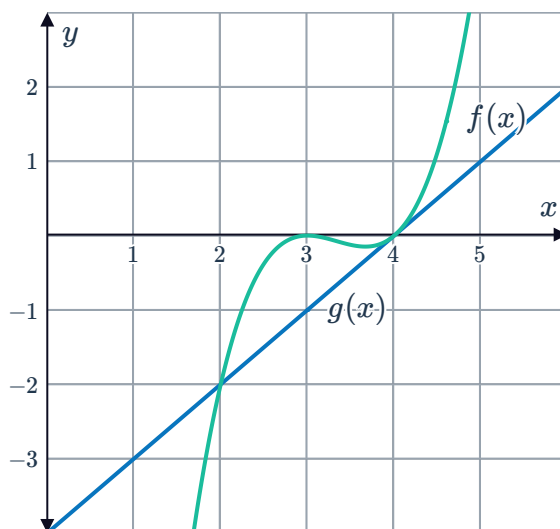
b



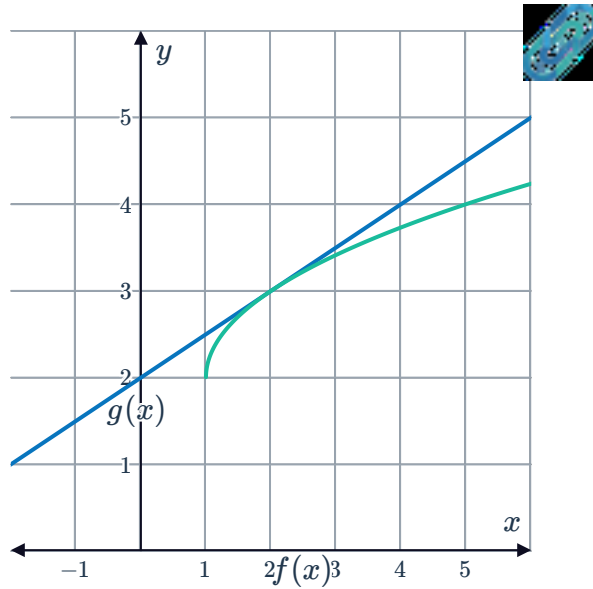
c



d



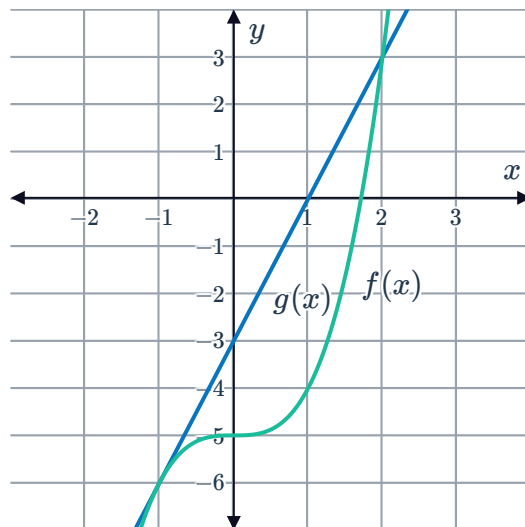
e



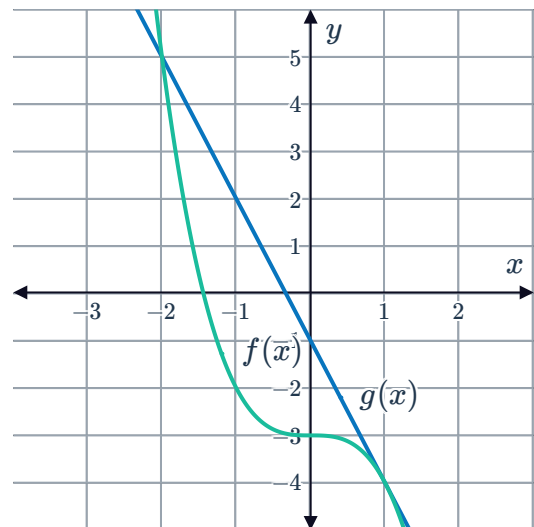
2 Each of the following graphs contain a curve, $f(x)$, along with one of its tangents, $g(x)$.

- State the coordinates of the point at which $g(x)$ is a tangent to the curve $f(x)$.
- State the gradient of the tangent.
- Hence determine the equation of the line $y = g(x)$.
- State the x -coordinate of the point on the curve at which we could draw a tangent that has the same gradient as $g(x)$.

a



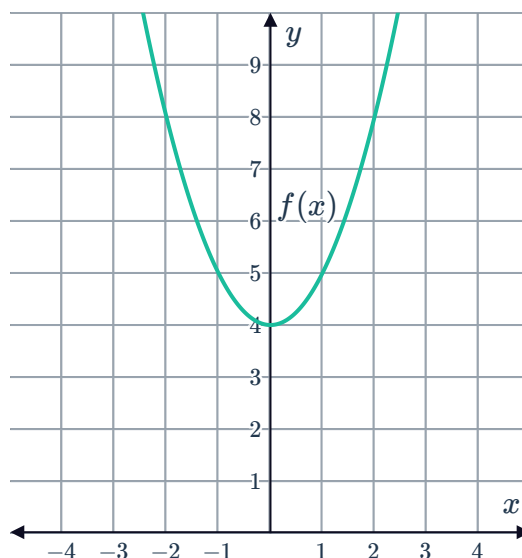
b



3 Consider the graph of the function $f(x)$:

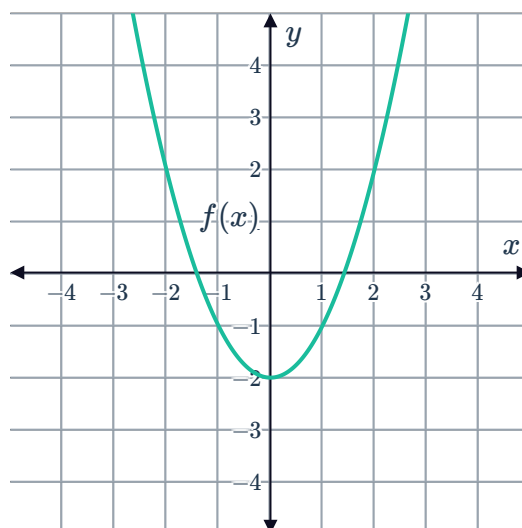


- a Sketch the graph of the function $g(x) = 2x + 3$ on the same number plane.
- b Is $g(x)$ a tangent to $f(x)$? Explain your answer.



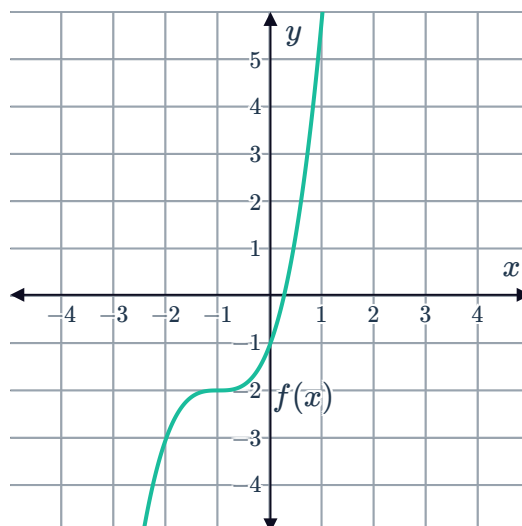
4 Consider the graph of the function $f(x)$:

- a Sketch the graph of the function $g(x) = 2x - 1$ on the same number plane.
- b Is $g(x)$ a tangent to $f(x)$? Explain your answer.



5 Consider the graph of the function $f(x)$:

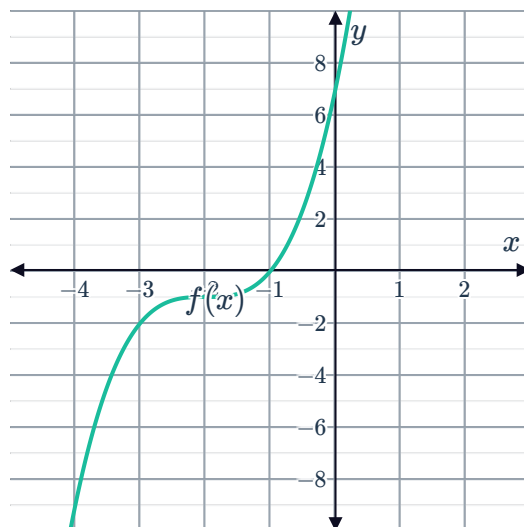
- a Sketch the graph of the function $g(x) = 3x + 3$ on the same number plane.
- b Is $g(x)$ a tangent to $f(x)$? Explain your answer.



6 Consider the graph of the function $f(x)$:



- a Sketch the graph of the function $g(x) = 4x + 7$ on the same number plane.
- b Is $g(x)$ a tangent to $f(x)$? Explain your answer.



7 Consider the curve given by the function $f(x) = x^2 - 1$.

- a Find the gradient of the tangent to the curve at the point $(1, 0)$.
- b Graph the curve and the tangent at the point $(1, 0)$ on a number plane.

8 Consider the curve given by the function $f(x) = x^2 - 4x + 2$.

- a Find the gradient of the tangent to the curve at the point $(3, -1)$.
- b State the coordinates of the vertex of the parabola $f(x) = x^2 - 4x + 2$.
- c Graph the curve and the tangent at the point $(3, -1)$ on a number plane.

Equation of a tangent at a given point

9 Consider the function $y = x^2 - 3x + 4$. State the x -coordinate of the point on the curve where the tangent makes an angle of 45° with the x -axis.

10 Consider the tangent to the curve $f(x) = x^2$ at the point $(-1, 1)$.

- a Find the gradient of the function $f(x) = x^2$ at this point.
- b Find the equation of the tangent to the curve $f(x) = x^2$ at the point $(-1, 1)$.

11 Consider the tangent to the curve $f(x) = x^3$ at the point $(-1, -1)$.

- a Find the gradient of the function $f(x) = x^3$ at this point.
- b Hence find the equation of the tangent to the curve $f(x) = x^3$ at the point $(-1, -1)$.

- 12 Consider the tangent to the curve $f(x) = -x^2$ at the point $(-3, -9)$.
- Find the gradient of the function $f(x) = -x^2$ at this point.
 - Hence find the equation of the tangent to the curve $f(x) = -x^2$ at the point $(-3, -9)$.
- 13 Consider the tangent to the curve $f(x) = x^2$ at the point $(-2, 4)$.
- Find the gradient of the function $f(x) = x^2$ at this point.
 - Find the equation of the tangent to the curve $f(x) = x^2$ at the point $(-2, 4)$.
- 14 Consider the function $f(x) = 3x^2$.
- Find the gradient of the function at $x = 2$.
 - Find the y -coordinate of the point on the function at $x = 2$.
 - Hence find the equation of the tangent to the curve $f(x) = 3x^2$ at $x = 2$.
- 15 Find the equation of the tangent to the curve $f(x) = 0.3x^3 - 5x^2 - x + 4$ at $x = 1$.

Applications

- 16 For each of the following:
- Find the x -coordinate of point M .
 - Find the y -coordinate of point M .
- At point $M(x, y)$, the equation of the tangent to the curve $y = x^2$ is given by $y = 4x - 4$.
 - At point $M(x, y)$, the equation of the tangent to the curve $y = x^3$ is given by $y = 12x - 16$.
- 17 Consider the function $f(x) = \frac{4x^3}{3} + \frac{5x^2}{2} - 3x + 7$. Find the x -coordinates of the points on the curve whose tangent is parallel to the line $y = 3x + 7$.
- 18 $5x + y + 2 = 0$ is the tangent to the curve $y = x^2 + bx + c$ at the point $(9, -47)$.
- Find the derivative $\frac{dy}{dx}$ of $y = x^2 + bx + c$.
 - Find the gradient of the tangent to the curve at $x = 9$.
 - Solve for the value of b .
 - Solve for the value of c .

19 Consider the function $f(x) = x^2 + 5x$.



- a Find the x -coordinate of the point at which $f(x)$ has a gradient of 13.
- b Hence state the coordinates of the point on the curve where the gradient is 13.

20 Consider the function $f(x) = x^3 - 6x^2$.

- a Find the x -coordinates of the points at which $f(x)$ has a gradient of 495.
- b Hence state the coordinates of the points on the curve where the gradient is 495.

21 Consider the function $f(x) = x^3 + 3x^2 - 19x + 2$.

- a Find the x -coordinates of the points at which $f(x)$ has a gradient of 5.
- b Hence state the coordinates of the points on the curve where the gradient is 5.

22 Consider the function $f(x) = x^3 + 6x^2 - 14x - 2$.

- a Find the x -coordinates of the points on the curve where the gradient is the same as that of $g(x) = 22x - 4$.
- b Hence state the coordinates of the points on the curve where the gradient is the same as that of $g(x) = 22x - 4$.

23 The curve $f(x) = k\sqrt{x} - 5x$ has a gradient of 0 at $x = 16$. Find the value of k .

24 Consider the function $y = 4x^2 - 5x + 2$.

- a Find $\frac{dy}{dx}$.
- b Hence find the value of x at which the tangent to the parabola is parallel to the x -axis.

25 Consider the function $f(x) = 5x^2 + \frac{4}{x} - 1$. The tangent to the curve at the point $(2, 21)$ makes an angle of θ with the x -axis. Find θ , correct to the nearest degree.

26 The curve $y = ax^3 + bx^2 + 2x - 17$ has a gradient of 58 at the point $(2, 31)$. Find the values of a and b .

27 The graph of $y = ax^3 + bx^2 + cx + d$ intersects the x -axis at $(2, 0)$, where it has a gradient of 36. It also intersects the y -axis at $y = -28$, where the tangent is parallel to the x -axis. Find the values of a, b, c and d .

28 In the following graph, the line $y = \frac{x}{10} + b$ is a tangent to the graph of $f(x) = 6\sqrt{x}$ at $x = a$.

- a** Find the value of a .
- b** Find the value of b .

